Matrices & Vectors

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Addition and scalar multiplication**

Let A be the matrix with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and B be the matrix with rows $\langle -1, 4\rangle$ and $\langle 0, 2\rangle$. Find $A + B$ and $3A$.

$A + B$ has rows $\langle 1, 5\rangle$ and $\langle 3, 1\rangle$. $3A$ has rows $\langle 6, 3\rangle$ and $\langle 9, -3\rangle$.

2 **[Foundational]** • **Target: Addition and scalar multiplication**

Using the same A and B, find $A - 2B$.

$A - 2B$ has rows $\langle 2 - (-2), 1 - 8\rangle = \langle 4, -7\rangle$ and $\langle 3 - 0, -1 - 4\rangle = \langle 3, -5\rangle$.

3 **[Foundational]** • **Target: Matrix multiplication**

Using A (rows $\langle 2, 1\rangle$, $\langle 3, -1\rangle$) and B (rows $\langle -1, 4\rangle$, $\langle 0, 2\rangle$) from above, find AB.

AB has rows $\langle -2, 10\rangle$ and $\langle -3, 10\rangle$: e.g. row 1, column 1 is $(2)(-1) + (1)(0) = -2$.

4 **[Foundational]** • **Target: Matrix multiplication**

Let A have rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and let $\mathbf{v} = \langle 2, -3\rangle$. Find $A\mathbf{v}$.

$A\mathbf{v} = \langle (2)(2) + (1)(-3), (3)(2) + (-1)(-3)\rangle = \langle 1, 9\rangle$.

5 **[Foundational]** • **Target: Matrix multiplication**

Explain, without computing BA, why AB and BA need not be equal for 2×2 matrices.

Matrix multiplication combines rows of the first factor with columns of the second; swapping the order pairs different rows with different columns, so in general $AB \neq BA$ (matrix multiplication is not commutative).

6 **[Foundational]** • **Target: Determinant and inverse**

Find $\det A$ for the matrix A with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and state whether A is invertible.

$\det A = (2)(-1) - (1)(3) = -5 \neq 0$, so A is invertible.

7 **[Foundational]** • **Target: Determinant and inverse**

Find A^{-1} for the matrix A with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$.

$A^{-1} = \frac{1}{-5} \times$ the matrix with rows $\langle -1, -1\rangle$ and $\langle -3, 2\rangle$, i.e., A^{-1} has rows $\langle 0.2, 0.2\rangle$ and $\langle 0.6, -0.4\rangle$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Determinant and inverse**

Without computing it, explain why the matrix with rows $\langle 4, 2\rangle$ and $\langle 6, 3\rangle$ has no inverse.

$\det = (4)(3) - (2)(6) = 12 - 12 = 0$, and a matrix with determinant 0 has no inverse (Theorem 2.1).

9 **[Intermediate]** • **Target: Solving systems with the inverse matrix**

Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

M has rows $\langle 3, -1\rangle$ and $\langle 1, 2\rangle$, so $\det M = (3)(2) - (-1)(1) = 7$, and M^{-1} has rows $\langle \frac{2}{7}, \frac{1}{7}\rangle$ and $\langle -\frac{1}{7}, \frac{3}{7}\rangle$. Then $x = \frac{1}{7}[2(8) + 1(5)] = 3$ and $y = \frac{1}{7}[-1(8) + 3(5)] = 1$. Check: $3(3) - 1 = 8$ and $3 + 2(1) = 5$.

10 **[Intermediate]** • **Target: Vector magnitude and dot product**

What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?

A zero dot product means $\mathbf{u}$ and $\mathbf{w}$ are perpendicular (orthogonal).

11 **[Intermediate]** • **Target: Angle between vectors**

Find the angle between $\mathbf{p} = \langle 1, 0\rangle$ and $\mathbf{q} = \langle 1, 1\rangle$.

$\mathbf{p} \cdot \mathbf{q} = 1$, $|\mathbf{p}| = 1$, $|\mathbf{q}| = \sqrt{2}$. $\cos\theta = \frac{1}{\sqrt{2}} \approx 0.7071 \Rightarrow \theta = 45^\circ$.

12 **[Intermediate]** • **Target: Angle between vectors**

Find the angle between $\langle 6, -8\rangle$ and $\langle 3, 4\rangle$, rounded to the nearest tenth of a degree.

Dot product $= 18 - 32 = -14$, magnitudes 10 and 5. $\cos\theta = \frac{-14}{50} = -0.28 \Rightarrow \theta \approx 106.3^\circ$.

13 **[Intermediate]** • **Target: Addition and scalar multiplication**

Given 3D vectors $\mathbf{u} = \langle 2, -3, 6\rangle$ and $\mathbf{v} = \langle 1, 4, 2\rangle$: (a) Find the unit vector in the direction of $\mathbf{u}$. (b) Calculate the angle θ between $\mathbf{u}$ and $\mathbf{v}$ to the nearest degree.

Part (a): Find the unit vector $\hat{\mathbf{u}}$: 1. Magnitude of $\mathbf{u}$: $|\mathbf{u}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. 2. Unit vector: $\hat{\mathbf{u}} = \frac{\mathbf{u}}{|\mathbf{u}|} = \left\langle \frac{2}{7}, -\frac{3}{7}, \frac{6}{7}\right\rangle$. Part (b): Find the angle θ: 1. Dot product $\mathbf{u} \cdot \mathbf{v}$: $\mathbf{u} \cdot \mathbf{v} = (2)(1) + (-3)(4) + (6)(2) = 2 - 12 + 12 = 2$. 2. Magnitude of $\mathbf{v}$.

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14 [Intermediate] • Target: Matrix multiplication

Find all real values of the constant k for which the matrix $M = (k+2, 4; 3, k-2)$ is singular (has no inverse).

Step 1: Set the determinant to zero: A matrix is singular if and only if $\det(M) = 0$: $\det(M) = (k+2)(k-2) - (4)(3) = 0$. Step 2: Solve the algebraic equation: $(k^2 - 4) - 12 = 0 \Rightarrow k^2 - 16 = 0$. Step 3: Extract roots: $(k-4)(k+4) = 0 \Rightarrow \mathbf{k = 4}$ or $\mathbf{k = -4}$. Check: For $k = 4$, $M = (6, 4; 3, 2)$, $\det = 12 - 12 = 0$ ✓. For $k = -4$, $M = (-2, 4; 3, -6)$, $\det = 12 - 12 = 0$ ✓.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Determinant and inverse • Cross-Lesson: 1.01 Complex Numbers

Find the magnitude and direction angle θ (in degrees, $0^\circ \le \theta < 360^\circ$) of the vector $\mathbf{v} = \langle -5, 5\sqrt{3} \rangle$.

Step 1: Compute magnitude: $|\mathbf{v}| = \sqrt{(-5)^2 + (5\sqrt{3})^2} = \sqrt{25 + 75} = \sqrt{100} = \mathbf{10}$. Step 2: Direction angle: Since $v_x = -5 < 0$ and $v_y = 5\sqrt{3} > 0$, the vector lies in Quadrant II. Reference angle: $\alpha = \tan^{-1}\left(\frac{5\sqrt{3}}{5}\right) = \tan^{-1}(\sqrt{3}) = 60^\circ$. $\theta = 180^\circ - 60^\circ = \mathbf{120}^\circ$.

16 [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 3.01 Sequences & Series

Find the angle θ between the vectors $\mathbf{a} = \langle 1, 2, 2 \rangle$ and $\mathbf{b} = \langle 2, -2, 1 \rangle$.

Step 1: Compute dot product: $\mathbf{a} \cdot \mathbf{b} = (1)(2) + (2)(-2) + (2)(1) = 2 - 4 + 2 = 0$. Step 2: Conclude angle: Since $\mathbf{a} \cdot \mathbf{b} = 0$ and both vectors have non-zero magnitude ($|\mathbf{a}| = 3$, $|\mathbf{b}| = 3$): $\cos\theta = 0 \Rightarrow \theta = \mathbf{90}^\circ$ $\left(\text{or } \frac{\pi}{2} \text{ rad}\right)$.

17 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 4.01 Advanced Functions & Equations

Find the inverse of the matrix $A = (4, 7; 1, 2)$.

Step 1: Compute determinant: $\det(A) = (4)(2) - (7)(1) = 8 - 7 = 1$. Step 2: Apply the inverse formula: $A^{-1} = \frac{1}{\det(A)}(d, -b; -c, a) = \frac{1}{1}(2, -7; -1, 4) = (\mathbf{2, -7; -1, 4})$. Check: $AA^{-1} = (8-7, -28+28; 2-2, -7+8) = (1, 0; 0, 1) = I$ ✓.

18 [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Compute the matrix product AB where $A = (1, 3; -2, 4)$ and $B = (2, 0; 5, -1)$.

Step 1: Compute rows of A dot columns of B: $(AB)_{11} = (1)(2) + (3)(5) = 2 + 15 = 17$ $(AB)_{12} = (1)(0) + (3)(-1) = 0 - 3 = -3$ $(AB)_{21} = (-2)(2) + (4)(5) = -4 + 20 = 16$ $(AB)_{22} = (-2)(0) + (4)(-1) = 0 - 4 = -4$ Conclusion: $AB = (\mathbf{17, -3; 16, -4})$.

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19 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 6.01 Limits

Vectors $\mathbf{a} = \langle 4, k \rangle$ and $\mathbf{b} = \langle 6, -8 \rangle$ are perpendicular. Find the value of k, and compute the projection of $\mathbf{c} = \langle 10, 5 \rangle$ onto $\mathbf{b}$.

Step 1: Use the orthogonality condition $\mathbf{a} \cdot \mathbf{b} = 0$: $\mathbf{a} \cdot \mathbf{b} = (4)(6) + (k)(-8) = 24 - 8k = 0 \Rightarrow 8k = 24 \Rightarrow \mathbf{k = 3}$. Step 2: Compute the vector projection of $\mathbf{c}$ onto $\mathbf{b}$: The scalar projection is $\text{comp}_{\mathbf{b}}\mathbf{c} = \frac{\mathbf{c} \cdot \mathbf{b}}{|\mathbf{b}|}$. 1. $\mathbf{c} \cdot \mathbf{b} = (10)(6) + (5)(-8) = 60 - 40 = 20$. 2. $|\mathbf{b}| = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$. 3. Vec.

20 [Challenging] • Target: Matrix multiplication • Cross-Lesson: 1.01 Complex Numbers

Let $A = (3, 1; 5, 2)$. Find A^{-1} and use it to solve the linear system: $3x + y = 7$ $5x + 2y = 12$

Step 1: Compute the determinant of A: $\det(A) = ad - bc = (3)(2) - (1)(5) = 6 - 5 = 1$. Step 2: Construct the inverse A^{-1}: Swap the main diagonal entries, negate the off-diagonal entries, and divide by $\det(A) = 1$: $A^{-1} = \frac{1}{1}(2, -1; -5, 3) = (\mathbf{2, -1; -5, 3})$. Step 3: Solve the matrix equation $A\mathbf{x} = \mathbf{b}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Addition and scalar multiplication

For $\mathbf{u} = \langle 20, 21 \rangle$ and $\mathbf{v} = \langle -5, -12 \rangle$, find $\cos\theta$ where θ is the angle between them, using $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{352}{377}$

b. $-\frac{352}{377}$ ✓

c. $-\frac{27}{29}$

d. $-\frac{176}{21}$

$\mathbf{u} \cdot \mathbf{v} = (20)(-5) + (21)(-12) = -352$. $|\mathbf{u}| = 29$, $|\mathbf{v}| = 13$. $\cos\theta = -352/377 = -\frac{352}{377}$.

22 [Foundational] • Target: Matrix multiplication

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the coordinate plane such that $|\mathbf{u}| = 5$, $|\mathbf{v}| = 10$, and the angle θ between them is obtuse ($90^\circ < \theta < 180^\circ$). If the minimum value of $|t\mathbf{u} + \mathbf{v}|$ over all real numbers t is equal to 6, what is the value of the dot product $\mathbf{u} \cdot \mathbf{v}$?

a. -50

b. **-40** ✓

c. -30

d. 40

We can solve this problem algebraically or geometrically. **Method 1: Algebraic (Quadratic Optimization)** Consider the square of the magnitude $|t\mathbf{u} + \mathbf{v}|$: $|t\mathbf{u} + \mathbf{v}|^2 = (t\mathbf{u} + \mathbf{v}) \cdot (t\mathbf{u} + \mathbf{v}) = t^2|\mathbf{u}|^2 + 2t(\mathbf{u} \cdot \mathbf{v}) + |\mathbf{v}|^2$ Let $D = \mathbf{u} \cdot \mathbf{v}$. Substituting $|\mathbf{u}| = 5$ and $|\mathbf{v}| = 10$: $f(t) = 25t^2 + 2Dt + 100$ This is a quadratic function in t with leading coefficien.

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23 [Foundational] • Target: Determinant and inverse

In the standard (x, y)-coordinate plane, vectors $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$ have magnitudes $||\mathbf{u}|| = 13$ and $||\mathbf{v}|| = 5$. If their dot product is $\mathbf{u} \cdot \mathbf{v} = 25$, and the matrix $M = [a, b; c, d]$ has $\det(M) > 0$, what is the value of $\det(M)$?

a. 25

b. 40

c. **60** ✓

d. 65

We can solve this using either Lagrange's algebraic identity or the geometric interpretation of the dot product and determinant. **Method 1: Lagrange's Identity** For any real numbers a, b, c, d, the identity connecting the determinant and the dot product is: $(ad - bc)^2 + (ac + bd)^2 = (a^2 + b^2)(c^2 + d^2)$ Notice that: - $\det(M) = ad - bc$ - $\mathbf{u} \cdot \mathbf{v} = ac + bd = 25$ - $||\mathbf{u}||^2 = a^2 + b^2 = 13^2 = 169$ - $||\mathbf{v}||^2 = c^2 + d^2 = 5^2 = 25$ Substituting these values into th.

24 [Foundational] • Target: Solving systems with the inverse matrix

For all real values of θ, the matrix product $[\cos\theta, \sin\theta][6, 2; 2, 3][\cos\theta; \sin\theta]$ evaluates to a 1×1 matrix $[k]$. What is the maximum possible value of k?

a. 6

b. **7** ✓

c. 9

d. 14

Step 1: Perform the matrix multiplication. First, multiply the 1×2 row vector by the 2×2 matrix: $[\cos\theta, \sin\theta][6, 2; 2, 3] = [6\cos\theta + 2\sin\theta, 2\cos\theta + 3\sin\theta]$ Next, multiply this result by the 2×1 column vector: $k = (6\cos\theta + 2\sin\theta)\cos\theta + (2\cos\theta + 3\sin\theta)\sin\theta$ $k = 6\cos^2\theta + 4\sin\theta\cos\theta + 3\sin^2\theta$ Step 2: Express the quadratic trigonometric terms using double-angle.

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25 [Foundational] • Target: Vector magnitude and dot product

Matrices A and B are defined as $A = [2, k; -1, 3]$ and $B = [1, 4; 2, k]$, where k is a real number. If the determinant of the matrix sum $A + B$ is equal to 17, what is the value of k? (Note: The determinant of a 2×2 matrix $[a, b; c, d]$ is defined as $ad - bc$.)

a. 1

b. 2

c. **6** ✓

d. $\frac{19}{2}$

Step 1: Compute the matrix sum $A + B$ by adding corresponding entries:

$A + B = [2 + 1, k + 4; -1 + 2, 3 + k] = [3, k + 4; 1, k + 3]$ Step 2: Calculate the determinant of $A + B$ using the definition $\det[a, b; c, d] = ad - bc$: $\det(A + B) = (3)(k + 3) - (1)(k + 4)$ Distribute carefully across both terms: $\det(A + B) = 3k + 9 - k - 4 = 2k + 5$ Step 3: Set the determinant equal to 17 and solve for k: $2k + 5 = 17 \Rightarrow 2k = 12 \Rightarrow k = 6$ Common Misconception.

26 [Foundational] • Target: Angle between vectors

For a 2×2 matrix $[a, b; c, d]$, the determinant is defined as $ad - bc$. Given the matrices $A = [x, 2; -1, 3]$ and $B = [1, -4; 3, x]$, the determinant of the matrix $A + B$ is equal to 19. If $x > 0$, what is the value of x?

a. **2** ✓

b. $\frac{5}{4}$

c. 4

d. 6

First, find the matrix sum $A + B$ by adding corresponding entries:

$A + B = [x + 1, 2 + (-4); -1 + 3, 3 + x] = [x + 1, -2; 2, x + 3]$ Next, calculate the determinant of $A + B$: $\det(A + B) = (x + 1)(x + 3) - (-2)(2)$ $\det(A + B) = (x^2 + 4x + 3) - (-4) = x^2 + 4x + 7$ Set the determinant equal to 19 and solve for x: $x^2 + 4x + 7 = 19$ $x^2 + 4x - 12 = 0$ Factoring the quadratic gives: $(x + 6)(x - 2) = 0$ This yields $x = -6$ or $x = 2$. Since we are given that $x > 0$, the value is.

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27 [Foundational] • Target: Addition and scalar multiplication

In the standard coordinate plane, vector $\mathbf{u} = \langle 2, -1 \rangle$ and vector $\mathbf{v} = \langle k, 5 \rangle$, where k is a real constant. If the magnitude of the vector $\mathbf{u} + \mathbf{v}$ is equal to 5, which of the following is a possible value of k?

a. **-5** ✓

b. -1

c. 3

d. 7

First, find the sum of vectors $\mathbf{u}$ and $\mathbf{v}$ by adding their corresponding components: $\mathbf{u} + \mathbf{v} = \langle 2 + k, -1 + 5 \rangle = \langle 2 + k, 4 \rangle$ Next, use the magnitude formula $||\langle x, y \rangle|| = \sqrt{x^2 + y^2}$: $\sqrt{(2 + k)^2 + 4^2} = 5$ Square both sides of the equation to eliminate the radical: $(2 + k)^2 + 16 = 25$ $(2 + k)^2 = 9$ Take the square root of both sides: $2 + k = 3 \quad \text{or} \quad 2 + k = -3$ $k = 1 \quad \text{or} \quad k = -5$ Amon.

28 [Foundational] • Target: Matrix multiplication

What is the value of the determinant of the matrix $[7, -3; 2, 4]$?

a. **34** ✓

b. 22

c. -29

d. -34

The determinant of a 2×2 matrix $[a, b; c, d]$ is given by the formula: $\det = ad - bc$ Substitute the entries $a = 7$, $b = -3$, $c = 2$, and $d = 4$: $\det = (7)(4) - (-3)(2) = 28 - (-6) = 28 + 6 = 34$ Common misconceptions: - Calculating $ad + bc$ gives $28 + (-6) = 22$. - Reversing the subtraction order as $bc - ad$ gives $-6 - 28 = -34$. - Multiplying along the rows instead of diagonals gives $(7)(-3) - (2)(4) = -21 - 8 = -29$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Determinant and inverse

Find the angle θ between $\mathbf{u} = \langle -6, 0 \rangle$ and $\mathbf{v} = \langle 6, -1 \rangle$, to the nearest 0.1°.

a. 170.5° ✓

b. 9.5°

c. 180.5°

d. 165.5°

$\mathbf{u} \cdot \mathbf{v} = -36$, $|\mathbf{u}| = \sqrt{36}$, $|\mathbf{v}| = \sqrt{37}$. $\cos\theta = \frac{-36}{\sqrt{36}\sqrt{37}} \approx -0.9864$, so $\theta \approx 170.5^\circ$ (Theorem 2.2).

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30 [Intermediate] • Target: Solving systems with the inverse matrix

Solve using the inverse-matrix method: $\{-5x + 3y = 28;\ -5x + 0y = 25$.

a. **x=-5,\ y=1** ✓

b. x=1,\ y=-5

c. x=-4,\ y=1

d. x=-5,\ y=-1

$\det A = 15$. Using A^{-1} (Theorem 2.1) on $\mathbf{b} = [28; 25]$ gives $x = -5,\ y = 1$. Check: $-5(-5) + 3(1) = 28$ ✓ and $-5(-5) + 0(1) = 25$ ✓.

31 [Intermediate] • Target: Vector magnitude and dot product

Given the vectors $\mathbf{u} = \langle 3,\ -4\rangle$ and $\mathbf{v} = \langle\ -2, 5\rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4,\ -3\rangle$ ✓

b. $\langle 1, 1\rangle$

c. $\langle 4, 1\rangle$

d. $\langle 8,\ -13\rangle$

To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$: $2\mathbf{u} = 2\langle 3,\ -4\rangle = \langle 2(3), 2(-4)\rangle = \langle 6,\ -8\rangle$ Next, add the corresponding components of $2\mathbf{u}$ and $\mathbf{v}$: $2\mathbf{u} + \mathbf{v} = \langle 6 + (-2),\ -8 + 5\rangle = \langle 4,\ -3\rangle$ - $\langle 1, 1\rangle$ is incorrect because it represents $\mathbf{u} + \mathbf{v}$, omitting the scalar factor of 2. - $\langle 4, 1\rangle$ is incorrect because the scal.

32 [Intermediate] • Target: Angle between vectors

If matrix $A = [3,\ -2; 0, 4]$ and matrix $B = [-1, 5; 2,\ -3]$, what is the matrix $2A + B$?

a. $[5, 1; 2, 5]$ ✓

b. $[2, 3; 2, 1]$

c. $[5, 3; 2, 1]$

d. $[7,\ -9;\ -2, 11]$

To find the matrix $2A + B$, first perform scalar multiplication by multiplying each entry of matrix A by 2: $2A = 2[3,\ -2; 0, 4] = [2(3), 2(-2); 2(0), 2(4)] = [6,\ -4; 0, 8]$ Next, add the corresponding entries of $2A$ and B:

$2A + B = [6,\ -4; 0, 8] + [-1, 5; 2,\ -3] = [6 + (-1),\ -4 + 5; 0 + 2, 8 + (-3)] = [5, 1; 2, 5]$

33 [Intermediate] • Target: Addition and scalar multiplication

Given matrix $A = (2,\ -1; 3, 3)$, what is the scalar product $3A$?

a. $(5, 2; 6, 6)$

b. $(6,\ -1; 3, 9)$

c. $(6,\ -3; 9, 9)$ ✓

d. $(6, 3; 9, 10)$

Multiply each element by 3: $3(2) = 6$, $3(-1) = -3$, $3(3) = 9$, $3(3) = 9$.

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34 [Intermediate] • Target: Matrix multiplication

A transformation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation can be represented by matrix multiplication $(x\ y)' = M(x\ y) + (a\ b)$. Find the values of a and b.

a. **a = -5, b = 0** ✓

b. a = 5, b = 0

c. a = 0, b = -5

d. a = 0, b = 5

Let the transformation be given by $(x\ y)' = M(x\ y) + (a\ b)$. We are given two points and their images under this transformation: 1. $A(1, 2) \to A'(-4, 2)$: $(-4\ 2) = M(1\ 2) + (a\ b)$ 2. $B(3, 5) \to B'(-2, 5)$: $(-2\ 5) = M(3\ 5) + (a\ b)$ Subtracting the first equation from the second, we get: $(-2 - (-4)\ 5 - 2) = M(3 - 1\ 5 - 2)$.

35 [Intermediate] • Target: Determinant and inverse

Consider a linear transformation T represented by the matrix $A = (2, 1\ 1, 3)$. Let $P = (x; y)$ be a point in the plane. The transformation maps P to P'. If the area of the region transformed by T from a unit square in the xy-plane is A, and the determinant of A is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is 5.

a. **5** ✓

b. 2

c. 1

d. 10

The area of the region transformed by a linear transformation T represented by a matrix A from a unit square is given by the absolute value of the determinant of A, i.e., Area $= |\det(A)|$. In this problem, we are given that the matrix representing the transformation is $A = (2, 1\ 1, 3)$. The determinant of A is calculated as: $\det(A) = (2 \times 3) - (1 \times 1) = 6 - 1 = 5$. We are also given that the area of the transformed region is 5. Thus, $|\det(A)| = 5$. Since we calculated.

36 [Intermediate] • Target: Solving systems with the inverse matrix

Consider the matrix $A = (2, 0; 0, 3)$. When the matrix A is applied to the vector $\mathbf{v} = (x; y)$, the resulting vector is $A\mathbf{v} = (6; 9)$. What is the original vector $\mathbf{v}$?

a. $(3; 3)$ ✓

b. $(1; 2)$

c. $(6; 3)$

d. $(12; 27)$

The matrix multiplication is given by: $(2, 0; 0, 3)(x; y) = (2x; 3y)$ We are given that this result is $(6; 9)$. Therefore, we have the system of equations: $2x = 6\ 3y = 9$ Solving for x and y: $x = \frac{6}{2} = 3\ y = \frac{9}{3} = 3$ So, the original vector $\mathbf{v}$ is $(3; 3)$.

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37 [Intermediate] • Target: Vector magnitude and dot product

Compute the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle 2, 5 \rangle$.

a. **19** ✓

b. 22

c. 16

d. 14

$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 = (2)(2) + (3)(5) = 4 + 15 = 19.$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

What is the magnitude of the 2D vector $\mathbf{v} = \langle 3, -4 \rangle$?

a. **5** ✓

b. 12

c. 10

d. 7

$||\mathbf{v}|| = \sqrt{(3)^2 + (-4)^2} = \sqrt{25} = 5.$

39 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 6.01 Limits

In ΔABC, the side lengths are $AB = 5$ and $BC = 4$, and $\angle B$ is an acute angle. If $\cos(2B) = -\frac{31}{32}$, what is the length of side AC?

a. 6 ✓

b. $\sqrt{46}$

c. $\sqrt{41}$

d. $\sqrt{31}$

We solve the problem in two steps by first determining $\cos(B)$ and then applying the Law of Cosines to find AC. **Step 1: Find $\cos(B)$** Using the double-angle identity for cosine: $\cos(2B) = 2\cos^2(B) - 1$ Substitute the given value $\cos(2B) = -\frac{31}{32}$: $-\frac{31}{32} = 2\cos^2(B) - 1$ $2\cos^2(B) = 1 - \frac{31}{32} = \frac{1}{32}$ $\cos^2(B) = \frac{1}{64}$ Since $\angle B$ is acute (0° 0: $\cos(B) = \frac{1}{8}$ **Step 2: Apply the Law of Cosines** In ΔABC, the side opposite.

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40 [Challenging] • Target: Matrix multiplication • Cross-Lesson: 1.01 Complex Numbers

In ΔABC, the lengths of sides AB and BC are 5 and 8, respectively, and the measure of $\angle B$ is 60°. What is the length of the altitude from vertex B to side AC?

a. $\frac{20\sqrt{3}}{7}$ ✓

b. $\frac{40\sqrt{3}}{7}$

c. $\frac{10\sqrt{3}}{7}$

d. $\frac{20}{7}$

To find the altitude from vertex B to side AC, we first determine the length of side AC using the Law of Cosines: $AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(\angle B)$ Substitute the given values: $AC^2 = 5^2 + 8^2 - 2(5)(8)\cos(60^\circ) = 25 + 64 - 80(\frac{1}{2}) = 89 - 40 = 49$ Taking the positive square root gives $AC = 7$. Next, calculate the area of ΔABC using the sine formula for triangle area.

41 [Challenging] • Target: Determinant and inverse • Cross-Lesson: 3.01 Sequences & Series

In ΔABC, the length of side AB is x, the length of side BC is $x + 3$, and the length of side AC is 7. If $\angle B = 60^\circ$, what is the value of x?

a. **5** ✓

b. 8

c. 4

d. 6

To find x, apply the Law of Cosines to ΔABC with angle B: $AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(B)$ Substitute the given values $AC = 7$, $AB = x$, $BC = x + 3$, and $\angle B = 60^\circ$: $7^2 = x^2 + (x + 3)^2 - 2x(x + 3)\cos(60^\circ)$ Recall that $\cos(60^\circ) = \frac{1}{2}$: $49 = x^2 + (x^2 + 6x + 9) - 2x(x + 3)\left(\frac{1}{2}\right)$ $49 = 2x^2 + 6x + 9 - x(x + 3)$ $49 = 2x^2 + 6x + 9 - x^2 - 3x$ $49 = x^2 + 3x + 9$ Rearrange into standard quadratic form: $x^2 + 3x - 40 = 0$ Factor the quadrat.

42 [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 4.01 Advanced Functions & Equations

In ΔABC, $A = 64^\circ$, $a = 18$, $b = 9$. Find angle B to the nearest tenth (assume B acute).

a. 26.7° ✓

b. 28.7°

c. 89.3°

d. 23.7°

$\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = \frac{9\sin 64^\circ}{18} \approx 0.4494 \Rightarrow B \approx 26.7^\circ$.

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43 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

In ΔABC, $a = 13$, $b = 9$, included angle $C = 60^\circ$. Find c to the nearest tenth.

a. **11.5** ✓

b. 12.5

c. 10.8

d. 22

$c^2 = 13^2 + 9^2 - 2(13)(9)\cos 60^\circ \approx 133.00$, so $c \approx 11.5$.

44 [Challenging] • Target: Angle between vectors • Cross-Lesson: 6.01 Limits

In ΔABC, $a = 15$, $b = 7$, included angle $C = 60^\circ$. Find c.

a. **13** ✓

b. 14

c. 9

d. 17

$c^2 = 15^2 + 7^2 - 2(15)(7)\cos 60^\circ = 169$, so $c = 13$.

45 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 1.01 Complex Numbers

A linear transformation in the 2D plane is represented by the matrix $M = (a, b; c, d)$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of a, b, c, and d, and then calculate the sum $a + b + c + d$.

a. **9** ✓

b. 11

c. 7

d. 5

Let the linear transformation matrix be $M = (a, b; c, d)$. The transformation of a point $\mathbf{x}$ to $\mathbf{x}'$ is given by $M\mathbf{x} = \mathbf{x}'$. For point $P(1, 2)$ mapped to $P'(4, 9)$: $(a, b; c, d)(1; 2) = (4; 9)$ This yields the equations: 1) $a + 2b = 4$ 2) $c + 2d = 9$ For point $Q(3, 1)$ mapped to $Q'(7, 12)$: $(a, b; c, d)(3; 1) = (7; 12)$ This yields the equations: 3) $3a + b = 7$ 4) $3c + d = 12$ We need.

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